Example 2.18

Can you reduce the following numbers to surds of same order :

(i) $\sqrt{3}$ (ii) $\sqrt[4]{3}$ (iii) $\sqrt[3]{3}$

Solution

(i) $\sqrt{3} = 3^{\frac{1}{2}}$
$= 3^{\frac{6}{12}}$
$= \sqrt[12]{3^6}$
$= \sqrt[12]{729}$

(ii) $\sqrt[4]{3} = 3^{\frac{1}{4}}$
$= 3^{\frac{3}{12}}$
$= \sqrt[12]{3^3}$
$= \sqrt[12]{27}$

(iii) $\sqrt[3]{3} = 3^{\frac{1}{3}}$
$= 3^{\frac{4}{12}}$
$= \sqrt[12]{3^4}$
$= \sqrt[12]{81}$

The last row has surds of same order.

Example 2.19

1. Express the surds in the simplest form: i) $\sqrt{8}$ ii) $\sqrt[3]{192}$
2. Show that $\sqrt[3]{7} > \sqrt[4]{5}$.

Solution

1. (i) $\sqrt{8} = \sqrt{4\times 2} = 2\sqrt{2}$

   (ii) $\sqrt[3]{192} = \sqrt[3]{4\times 4\times 4\times 3} = 4\sqrt[3]{3}$

2. $\sqrt[3]{7} = \sqrt[12]{7^4} = \sqrt[12]{2401}$

   $\sqrt[4]{5} = 5^{\frac{1}{4}} = 5^{\frac{3}{12}} = \sqrt[12]{5^3} = \sqrt[12]{125}$

   $\sqrt[12]{2401} > \sqrt[12]{125}$

   Therefore, $\sqrt[3]{7} > \sqrt[4]{5}$.

(iii) Pure and Mixed Surds : A surd is called a pure surd if its coefficient in its simplest form is 1. For example, $\sqrt{3}$, $\sqrt[3]{6}$, $\sqrt[4]{7}$, $\sqrt[5]{49}$ are pure surds. A surd is called a mixed surd if its co-efficient in its simplest form is other than 1. For example, $5\sqrt{3}$, $2\sqrt[4]{5}$, $3\sqrt[4]{6}$ are mixed surds.

(iv) Simple and Compound Surds : A surd with a single term is said to be a simple surd. For example, $\sqrt{3}$, $2\sqrt{5}$ are simple surds. The algebraic sum of two (or more) surds is called a compound surd. For example, $\sqrt{5}+3\sqrt{2}, \sqrt{3}-2\sqrt{7}, \sqrt{5}-7\sqrt{2}+6\sqrt{3}$ are compound surds.

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